Telecom Customer Churn & Retention Analytics

1. Project Overview

This project analyzes customer churn in the telecommunications industry using Python, Pandas, MySQL, and Power BI.

Customer churn occurs when customers stop using a company's services. Understanding churn patterns helps businesses identify customer segments that may require better onboarding, service improvements, and retention strategies.

The project follows an end-to-end data analytics workflow, including data cleaning, exploratory data analysis (EDA), SQL analysis, interactive dashboard development, and business recommendations.

2. Project Objectives

The main objectives are:

- Clean and validate the telecom customer dataset.
- Calculate customer churn KPIs.
- Analyze churn across contract types, tenure groups, payment methods, internet services, and technical support.
- Use SQL to answer business questions.
- Build an interactive Power BI dashboard.
- Communicate findings through actionable business recommendations.

3. Tools and Technologies

Tool	Purpose
Python	Data preparation and analysis
Pandas	Data cleaning and manipulation
Jupyter Notebook	Analysis workflow
Matplotlib	Data visualization
MySQL	SQL queries and customer segmentation
Power BI Desktop	Interactive dashboard
Markdown	Project documentation

4. Dataset Description

The project uses IBM's public Telco Customer Churn sample dataset.

Dataset source: IBM Telco Customer Churn CSV

The original dataset contains 7,043 customer records and 21 columns.

Important fields include:

- `customerID` — unique customer identifier
- `gender` — customer gender
- `SeniorCitizen` — senior-citizen indicator
- `tenure` — number of months with the company
- `Contract` — contract type
- `PaymentMethod` — payment method
- `InternetService` — internet service type
- `TechSupport` — technical support subscription
- `MonthlyCharges` — monthly customer charges
- `TotalCharges` — total customer charges
- `Churn` — whether the customer left the company

5. Data Cleaning

Data cleaning was performed in `notebooks/01_Data_Cleaning.ipynb` using Python and Pandas.

The steps included:

1. Loading and inspecting the raw CSV dataset.
2. Checking column names, data types, and missing values.
3. Checking for duplicate records and duplicate customer IDs.
4. Identifying 11 blank values in `TotalCharges`.
5. Converting `TotalCharges` to a numeric data type.
6. Replacing blank `TotalCharges` values with zero for customers whose tenure was zero.
7. Removing unnecessary leading and trailing whitespace from text fields.
8. Saving the cleaned dataset as a CSV file.

Cleaning Validation

Validation	Result
Original records	7,043
Cleaned records	7,043
Columns	21
Duplicate customer IDs	0
Missing values after cleaning	0
Blank `TotalCharges` values after cleaning	0

All 7,043 records were preserved during cleaning.

6. Key Performance Indicators

The analysis produced the following overall metrics:

KPI	Result
Total Customers	7,043
Churned Customers	1,869
Retained Customers	5,174
Overall Churn Rate	26.54%

The churn rate was calculated as:

Churn Rate = (Churned Customers / Total Customers) × 100

The resulting churn rate indicates that approximately 26.54% of the customers in this dataset had churned.

7. Exploratory Data Analysis

Exploratory analysis examined how churn varies across customer groups.

7.1 Churn by Contract Type

Contract Type	Total Customers	Churned Customers	Churn Rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

Finding: Month-to-month customers have the highest observed churn rate, while customers on two-year contracts have the lowest.

7.2 Churn by Customer Tenure

Tenure Group	Total Customers	Churned Customers	Churn Rate
0–12 months	2,186	1,037	47.44%
13–24 months	1,024	294	28.71%
25–48 months	1,594	325	20.39%
49–72 months	2,239	213	9.51%

Finding: Customers in their first year have the highest observed churn rate. Churn rates decrease across the longer-tenure groups.

7.3 Churn by Payment Method

Payment Method	Total Customers	Churned Customers	Churn Rate
Electronic check	2,365	1,071	45.29%
Mailed check	1,612	308	19.11%

Payment Method	Total Customers	Churned Customers	Churn Rate
Bank transfer (automatic)	1,544	258	16.71%
Credit card (automatic)	1,522	232	15.24%

Finding: Electronic-check users have the highest observed churn rate among the payment-method groups.

7.4 Churn by Internet Service

Internet Service	Total Customers	Churned Customers	Churn Rate
Fiber optic	3,096	1,297	41.89%
DSL	2,421	459	18.96%
No internet service	1,526	113	7.40%

Finding: Fiber-optic customers have the highest observed churn rate among the internet-service groups.

7.5 Churn by Technical Support

Technical Support	Churn Rate
No	41.64%
Yes	15.17%
No internet service	7.40%

Finding: Customers without technical support have a higher observed churn rate than customers with support.

7.6 Churned vs. Retained Customers

Metric	Retained Customers	Churned Customers
Average tenure	37.57 months	17.98 months
Median tenure	38 months	10 months
Average monthly charges	61.27	74.44
Average total charges	2,549.91	1,531.80

Finding: Churned customers have a shorter average tenure and higher average monthly charges. Their lower average total charges are consistent with their shorter time as customers.

8. SQL Analysis

MySQL was used to independently validate key findings and segment customer churn.

The SQL analysis included:

- Overall churn distribution and customer percentages.

- Churn rates by contract type.
- Churn rates by tenure group.
- Churn rates by payment method.
- Churn rates by internet service.

The queries use SQL aggregations, `COUNT`, conditional `SUM`, `CASE WHEN`, `GROUP BY`, and percentage calculations.

The principal results from SQL aligned with those calculated in Python.

9. Power BI Dashboard

The interactive dashboard was created in Power BI Desktop and saved as:

`dashboard/Telco_Customer_Churn_Analytics.pbix`

The dashboard includes:

- Total Customers KPI card
- Churned Customers KPI card
- Churn Rate KPI card
- Churn Rate by Contract Type chart
- Churn Rate by Tenure Group chart
- Churn Rate by Payment Method chart
- Churn Rate by Internet Service chart
- Churn Rate by Technical Support chart
- Interactive Contract slicer

The slicer allows users to select contract types and explore how the selected group affects report metrics and charts.

10. Business Insights

The analysis highlights several patterns that are relevant to customer retention:

1. Month-to-month customers have a higher churn rate than customers on longer contracts.
2. Customers in their first year have the highest observed churn rate.
3. Electronic-check users have the highest churn rate among the payment methods analyzed.
4. Fiber-optic customers have a higher observed churn rate than DSL customers.
5. Customers without technical support have a higher observed churn rate than customers with support.
6. Churned customers have shorter average tenure and higher average monthly charges than retained customers.

These results identify customer segments that could benefit from further investigation.

11. Business Recommendations

Based on the observed patterns, the following actions could be evaluated:

- Improve onboarding and early engagement for new customers.
- Investigate retention offers and contract options for month-to-month customers.

- Review billing experience and payment preferences among electronic-check users.
- Investigate pricing, service quality, and customer expectations among fiber-optic users.
- Evaluate customer support availability and resolution quality.
- Monitor churn KPIs regularly to assess whether retention initiatives are associated with improvements.

These are recommendations for further investigation rather than proof that any single factor causes churn.

12. Project Files

The project is organized as follows:

```
telco_customer_churn_project/
├── dashboard/
│   └── Telco_Customer_Churn_Analytics.pbix
├── data/
│   ├── raw/
│   │   └── Telco-Customer-Churn.csv
│   └── processed/
│       └── Telco-Customer-Churn-Cleaned.csv
├── notebooks/
│   └── 01_Data_Cleaning.ipynb
├── report/
│   └── Telco_Customer_Churn_Analysis_Report.md
├── sql/
│   └── telco_churn_analysis.sql
└── README.md
```

13. Conclusion

This project demonstrates an end-to-end data analytics workflow, from data cleaning and exploratory analysis to SQL validation and interactive Power BI reporting.

It showcases practical skills in Python, Pandas, MySQL, data visualization, KPI development, dashboard design, and communicating business insights.

The results provide a starting point for further investigation into customer retention strategies.

14. Disclaimer

This project uses a public sample dataset for educational and portfolio purposes. The findings reflect the dataset analyzed and should not automatically be generalized to all telecommunications businesses.

The observed relationships are descriptive and do not independently establish causal factors behind customer churn.